

Grade 11 - Term 3

C1–C4 Practice Exercises and Solutions

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Evaluated criteria

- C1** Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.
- C2** Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.
- C3** Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.
- C4** Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution

must be identified from the characteristics of the situation.

C1 For each context, characterize the random variable and identify the appropriate probability distribution. [5]

Context 1. A university sends an invitation to exactly 60 randomly selected alumni to participate in a fundraising campaign. Each alumnus either makes a donation or does not. The university records the number of alumni who donate to estimate the expected response to the campaign.

Context 2. A courier company models the travel time for a delivery route. Travel times range from 25 to 55 minutes, with times around 40 minutes occurring most frequently. Values become progressively less common as they move away from 40 minutes toward either endpoint. The company uses this model to estimate delivery durations.

Context 3. A telecommunications company counts the number of incoming technical-support requests received by a service center during a fixed 30-minute period. Historical records indicate that requests occur independently and without a predictable pattern. The company uses this model to plan the number of available support agents.

Context 4. A water-utility company studies the daily water consumption of a small residential building. Consumption ranges from 100 to 500 litres, and the probability density increases steadily as consumption approaches 500 litres. The company uses this model to plan daily supply capacity.

Context 5. A manufacturer examines batches of circuit boards until the first board that fails a functional test is found. Boards are tested independently, and testing stops immediately after the first failure. The manufacturer records the number of boards tested, including the failed board, to estimate inspection requirements.

Context 6. A logistics company studies the delay, in minutes, of trucks arriving at a distribution center. Delays can take any value between 0 and 30 minutes, with all values in this interval considered equally likely. The company uses this model to estimate scheduling margins.

Context 7. A hospital records the number of patients arriving at an emergency department during each fixed four-hour shift. Comparable shifts show that arrivals occur independently and without a predictable timetable. The hospital uses this model to plan staffing levels.

Context 8. A cinema chain sends a discount code to exactly 150 randomly selected customers. Each customer either uses the code or does not. The chain records the number of customers who use the discount to estimate the additional attendance generated by the campaign.

Context 9. A dairy producer measures the fat content of milk samples from its regular production process. Measurements cluster around 3.5%. Measurements farther from 3.5% occur progressively less often. The producer uses this model to establish quality-control limits.

Context 10. A city's electric utility studies the daily electricity consumption of a small commercial building. Consumption ranges from 200 to 800 kWh, and the probability density decreases steadily as consumption moves from 200 kWh toward 800 kWh. The utility uses this model to estimate the likelihood of different daily loads.

C1 - Distribution identification: Solutions [5]

1. **Context 1 — Discrete, Binomial.** X_1 counts donations among 60 selected alumni, with each alumnus having two possible outcomes.
2. **Context 2 — Continuous, Triangular.** X_2 has a minimum of 25 minutes, maximum of 55 minutes, and mode of 40 minutes.
3. **Context 3 — Discrete, Poisson.** X_3 counts independent technical-support requests occurring during a fixed

time interval.

4. **Context 4 — Continuous, Linear.** X_4 has a density that increases steadily toward 500 litres.
5. **Context 5 — Discrete, Geometric.** X_5 counts the number of boards tested until the first failure.
6. **Context 6 — Continuous, Uniform.** X_6 can take any value in $[0, 30]$, with all values equally likely.
7. **Context 7 — Discrete, Poisson.** X_7 counts independent patient arrivals during a fixed period.
8. **Context 8 — Discrete, Binomial.** X_8 counts discount-code users among 150 customers, with each customer having two possible outcomes.
9. **Context 9 — Continuous, Normal.** X_9 is approximately symmetric and concentrated around 3.5%.
10. **Context 10 — Continuous, Linear.** X_{10} has a density that decreases steadily from 200 kWh toward 800 kWh.



2 Charging Time for an Electric Vehicle

[C2-3]

A Colombian transportation company studies the additional charging time required for electric vehicles at one of its charging stations. For a particular type of vehicle, the additional charging time can range from 0 to 4 minutes. A charging time of any length within this interval is possible, and shorter additional times become steadily more common. Historical data indicate that an additional charging time of 0 minutes is twice as common as an additional charging time of 4 minutes.

Questions/Tasks.

C2 Determine the probability that the additional charging time is between 1 and 3 minutes.

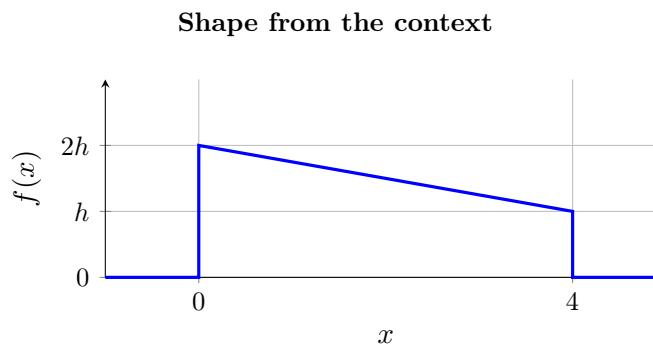
[3]

C2 - Linear probability distribution

[3]

2.1 Support and Shape

The random variable X has support $[0, 4]$. The density decreases linearly, and its height at 0 is twice its height at 4. Let the endpoint heights be $2h$ and h .



2.2 Total Area = 1

The total area under a probability density function is 1. For a trapezoid with parallel sides of heights h_1 and h_2 and base b ,

$$A = \frac{h_1 + h_2}{2}(b)$$

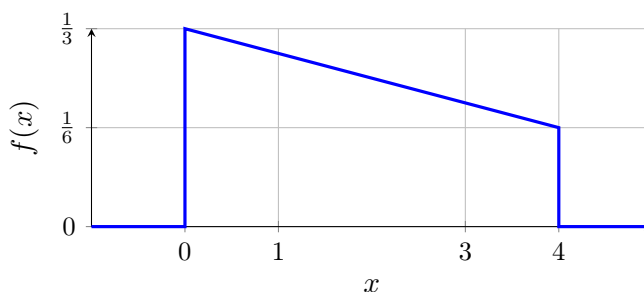
Therefore, in this case,

$$1 = \frac{2h + h}{2}(4) \quad \rightarrow \quad 1 = 3h(2) \quad \rightarrow \quad h = \frac{1}{6}.$$

Therefore,

$$f(0) = 2h = \frac{1}{3}, \quad f(4) = h = \frac{1}{6}.$$

Completed density graph



2.3 Complete pdf

The density is the straight line through

$$\begin{array}{cc} (x_1, y_1) & (x_2, y_2) \\ \left(0, \frac{1}{3}\right) & \left(4, \frac{1}{6}\right) \end{array}$$

For a general straight line,

$$f(x) = mx + b \quad \text{where} \quad m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{and} \quad b = y_1 - mx_1$$

Substituting the given points,

$$m = \frac{\frac{1}{6} - \frac{1}{3}}{4 - 0} = -\frac{1}{24} \quad b = \frac{1}{3} - \frac{-1}{24}(0) = \frac{1}{3}$$

$$f(x) = -\frac{x}{24} + \frac{1}{3} = -\frac{x}{24} + \frac{8}{24} = \frac{8 - x}{24}$$

Therefore,

$$f(x) = \begin{cases} \frac{8 - x}{24}, & 0 \leq x \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

2.4 Probability Between 1 and 3 Minutes

The required probability is the area under the density from 1 to 3:

$$P(1 \leq X \leq 3) = \frac{f(1) + f(3)}{2}(3 - 1)$$

Since

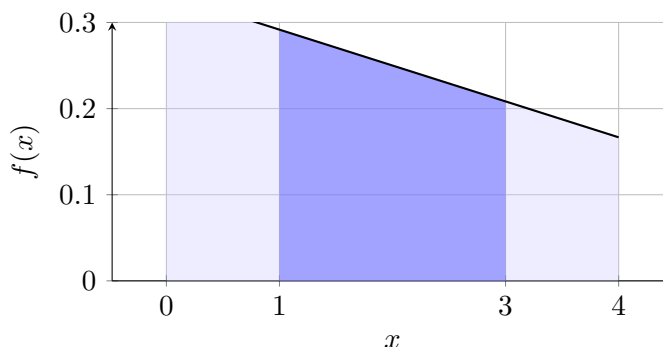
$$f(1) = \frac{8-1}{24} = \frac{7}{24}, \quad f(3) = \frac{8-3}{24} = \frac{5}{24}.$$

$$P(1 \leq X \leq 3) = \frac{1}{2} \left(\frac{1}{4} + \frac{1}{8} \right) (2) = \frac{3}{8}.$$

Therefore,

$$P(1 \leq X \leq 3) = \frac{3}{8} = 0.375 .$$

Band graph: $P(1 \leq X \leq 3)$



3 Daily Sales Revenue of a Local Business

[C2-4]

A local business records its daily sales revenue over several months. Daily revenue ranges from \$1 million to \$7 million. Revenue becomes steadily more common from \$1 million up to a most common level of \$6 million, then steadily less common through \$7 million; there are no observations exactly at either limit.

Questions/Tasks.

C2 Determine the probability that the daily sales revenue is between \$2 million and \$6 million.

[4]

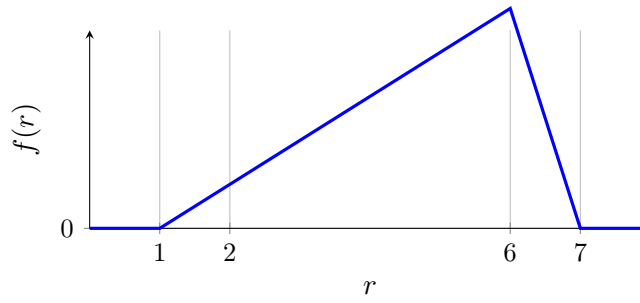
C2 - Triangular probability distribution

[4]

3.1 Support and Shape

Let R represent the daily sales revenue, measured in millions of dollars. The random variable R has support $[1, 7]$. Its density rises linearly to a peak at $r = 6$, then falls linearly to zero at $r = 7$. Let the peak height be H .

Shape from the context



3.2 Total Area = 1

The total area under a probability density function is 1. For a triangle with base b and height H ,

$$A = \frac{1}{2}bH.$$

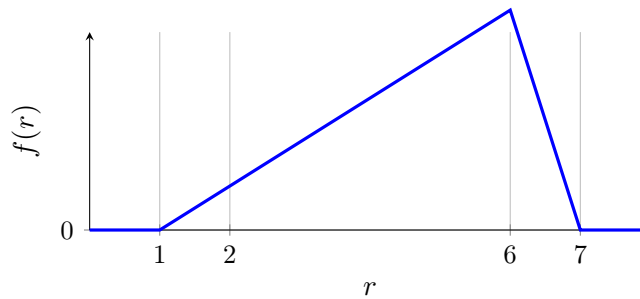
Therefore, in this case,

$$1 = \frac{1}{2}(7 - 1)H \quad \rightarrow \quad 1 = \frac{1}{2}(6)H \quad \rightarrow \quad 1 = 3H \quad \rightarrow \quad H = \frac{1}{3}.$$

Therefore,

$$f(6) = \frac{1}{3}.$$

Completed density graph



3.3 Complete pdf

The density consists of two straight-line segments. For a general straight line,

$$f(r) = mr + b, \quad \text{where} \quad m = \frac{y_2 - y_1}{x_2 - x_1} \quad \text{and} \quad b = y_1 - mx_1.$$

For the rising segment, the line passes through

$$\begin{array}{cc} (x_1, y_1) & (x_2, y_2) \\ (1, 0) & \left(6, \frac{1}{3}\right) \end{array}$$

Thus

$$m_1 = \frac{\frac{1}{3} - 0}{6 - 1} = \frac{1}{15} \quad b_1 = 0 - \frac{1}{15}(1) = -\frac{1}{15}.$$

Therefore,

$$f(r) = \frac{r}{15} - \frac{1}{15} = \frac{r - 1}{15}.$$

For the falling segment, the line passes through

$$\left(6, \frac{1}{3}\right) \quad \text{and} \quad (7, 0).$$

However, these calculations are not needed for the requested probability, because the interval $2 \leq R \leq 6$ ends at the mode. Therefore, the equation of the falling segment is stated directly without showing its derivation.

The complete probability density function is

$$f(r) = \begin{cases} \frac{r - 1}{15}, & 1 \leq r \leq 6, \\ \frac{7 - r}{3}, & 6 < r \leq 7, \\ 0, & \text{otherwise.} \end{cases}$$

3.4 Probability Between \$2 and \$6 Million

The complete triangular density has a base extending from $r = 1$ to $r = 7$. The required interval extends from $r = 2$ to the mode at $r = 6$. Therefore, the area outside the required interval consists of two tails: the left tail from $r = 1$ to $r = 2$, and the right tail from $r = 6$ to $r = 7$:

$$P(2 \leq R \leq 6) = 1 - P(R < 2) - P(R > 6).$$

For the left tail, the density is

$$f(r) = \frac{r - 1}{15}.$$

Thus,

$$f(2) = \frac{2 - 1}{15} = \frac{1}{15}.$$

The left-tail area is therefore

$$P(R < 2) = \frac{1}{2}(2 - 1) \left(\frac{1}{15} \right) = \frac{1}{30}.$$

At the mode the height is the height of the triangle as seen in part 1.

$$f(6) = \frac{1}{3}$$

The right-tail area is therefore

$$P(R > 6) = \frac{1}{2}(7 - 6) \left(\frac{1}{3} \right) = \frac{1}{6}.$$

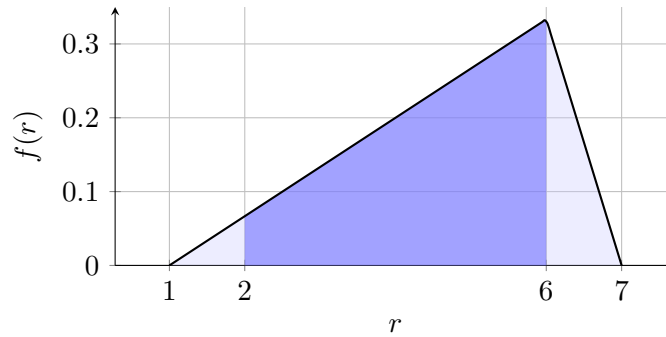
Consequently,

$$\begin{aligned}
 P(2 \leq R \leq 6) &= 1 - \frac{1}{30} - \frac{1}{6} \\
 &= 1 - \frac{1}{30} - \frac{5}{30} \\
 &= 1 - \frac{6}{30} \\
 &= \frac{24}{30} \\
 &= \frac{4}{5}.
 \end{aligned}$$

Therefore,

$$P(2 \leq R \leq 6) = \frac{4}{5} = 80\% .$$

Band graph: $P(2 \leq R \leq 6)$



4 Timing of an Overnight Price Adjustment

[C3-2]

A retail company monitors the time of day at which an unexpected price adjustment occurs in one of its online markets. The adjustment is equally likely to occur at any time during a 24-hour period, so no part of the day is more likely than another.

Questions/Tasks.

C3 Determine the probability that the price adjustment occurs between 8 p.m. and 6 a.m.

[2]

C3 - Uniform probability distribution

[2]

4.1 Uniform Distribution

Let T represent the time of the price adjustment, measured in hours after midnight. Since every time during the 24-hour period is equally likely,

$$T \sim U(0, 24) \quad \longrightarrow \quad f(t) = \frac{1}{24 - 0} = \frac{1}{24}.$$

4.2 Probability Between 8 p.m. and 6 a.m.

The interval crosses midnight, so it consists of

$$20 \leq T < 24 \quad \text{and} \quad 0 \leq T \leq 6.$$

The probability is found by calculating the area of each rectangle individually and then adding them:

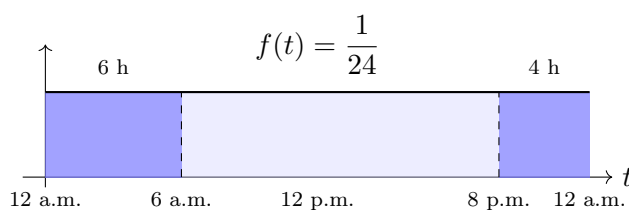
$$A_1 = (6 - 0) \left(\frac{1}{24} \right) = \frac{6}{24},$$

$$A_2 = (24 - 20) \left(\frac{1}{24} \right) = \frac{4}{24}.$$

Therefore,

$$P(\text{between 8 p.m. and 6 a.m.}) = A_1 + A_2 = \frac{6}{24} + \frac{4}{24} = \frac{10}{24} = \frac{5}{12} \approx 41.67\%.$$

Probability between 8 p.m. and 6 a.m.



5 Direction of an Unexpected Sensor Reading

[C3-3]

A directional sensor records the direction in which it is pointing. The sensor can point in any direction around a complete circle, and every direction is equally likely. Let A represent the direction of the sensor, measured in radians from a fixed reference marker at 0 radians.

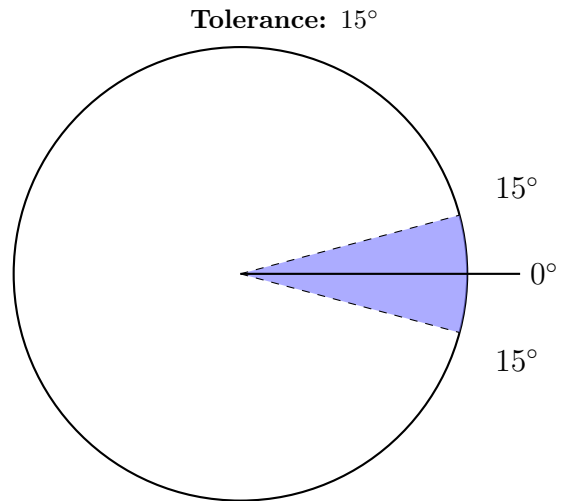
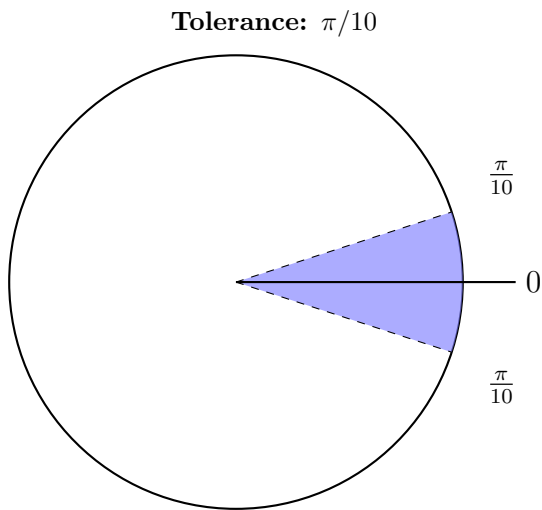
Questions/Tasks.

- C3** Determine which is more probable and by how much: the sensor pointing within $\pi/10$ radians of the reference marker, or the sensor pointing within 15° of the reference marker. [3]

C3 - Uniform probability distribution

[3]

Directions within a range around the reference marker

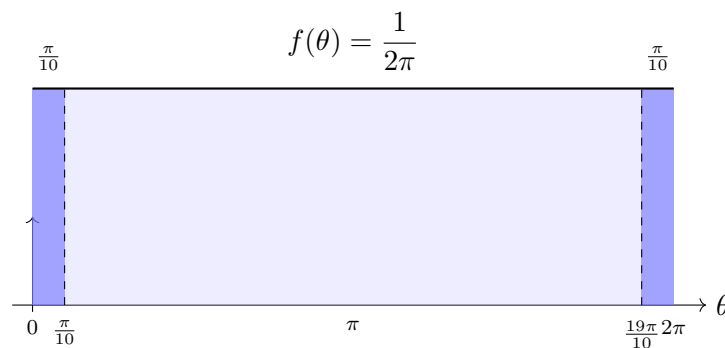


5.1 Probability Within $\pi/10$ of the Reference Marker

The sensor can point uniformly around a complete circle of 2π radians. Being within $\pi/10$ radians of the reference marker includes the same angular distance on both sides of the marker, so the total angular width is

$$2\left(\frac{\pi}{10}\right) = \frac{\pi}{5}. \quad \longrightarrow \quad P\left(\text{within } \frac{\pi}{10} \text{ of the reference marker}\right) = \frac{\pi/5}{2\pi} = \frac{1}{10} = 10\% .$$

Probability Within $\pi/10$ of the Reference Marker

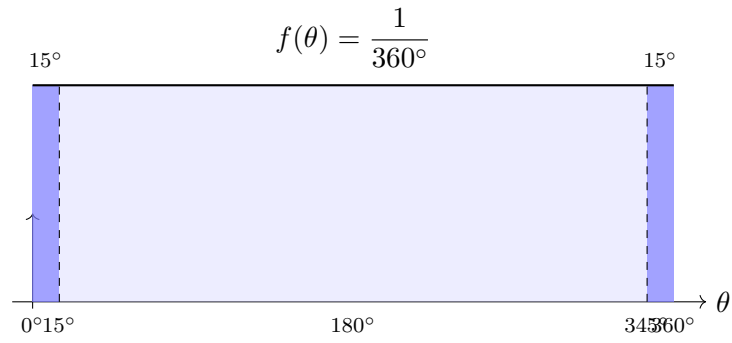


5.2 Probability Within 15° of the Reference Marker

A complete circle measures 360° . Being within 15° of the reference marker gives a total angular width of

$$2(15^\circ) = 30^\circ. \quad \longrightarrow \quad P(\text{within } 15^\circ \text{ of the reference marker}) = \frac{30^\circ}{360^\circ} = \frac{1}{12} \approx 8.33\% .$$

Probability Within 15° of the Reference Marker



5.3 Comparison

Since

$$\frac{1}{10} > \frac{1}{12}, \quad \longrightarrow \quad \frac{1}{10} - \frac{1}{12} = \frac{1}{60} \approx 1.67\%.$$

Thus,

Within $\frac{\pi}{10}$ radians is more probable than within 15° by $\frac{1}{60} \approx 1.67\%$.

6 Waiting for a Food-Delivery Driver [C4-0.5]

A food-delivery platform receives driver arrivals at an average rate of 5 drivers every 8 minutes.

Questions/Tasks.

C0.5 Determine the probability that the next driver arrives more than 2 minutes after the start of the observation. [0.5]

C4 - Exponential probability distribution [0.5]

The average rate is 5 drivers every 8 minutes, so

$$\lambda = \frac{5}{8} = 0.625.$$

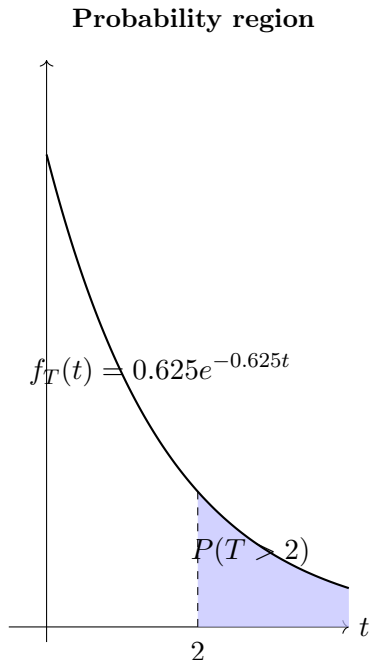
General case: $P(T > t) = e^{-\lambda t}.$

Thus,

$$P(T > 2) = e^{-(0.625)(2)} = e^{-1.25} \approx 0.2865.$$

Therefore,

$$P(T > 2) \approx 28.65\%.$$



7 Waiting for a Public-Transport Delay Alert [C4-0.5]

A public-transport monitoring system detects significant service delays at an average rate of 4 alerts every 15 minutes.

Questions/Tasks.

C0.5 Determine the probability that the next significant delay is detected within 3 minutes after the start of the observation. [0.5]

C4 - Exponential probability distribution [0.5]

The average rate is 4 significant delays every 15 minutes, so

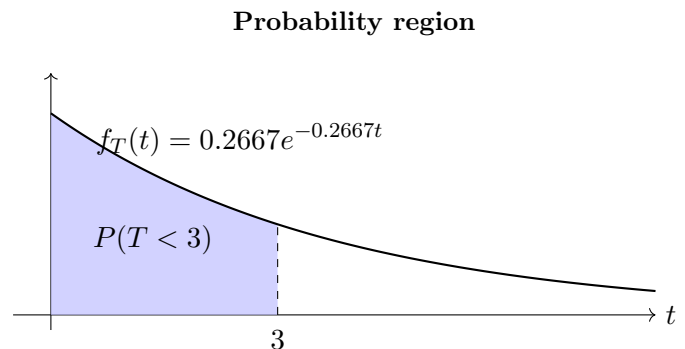
$$\lambda = \frac{4}{15} \approx 0.2667.$$

General case: $P(T < t) = 1 - e^{-\lambda t}.$

$$P(T < 3) = 1 - e^{-(0.2667)(3)} \approx 1 - e^{-0.8} \approx 0.5507.$$

Therefore,

$$P(T < 3) \approx 55.07\%.$$



8 Waiting for a Supply Chain Disruption Alert

[C4-1]

A supply chain monitoring system detects significant disruptions at an average rate of 5 alerts every 20 minutes.

Questions/Tasks.

- C1** Determine the probability that the next significant disruption is detected between 4 and 8 minutes after the observation time. [1]

C4 - Exponential probability distribution

[1]

The average rate is 5 significant disruptions every 20 minutes, so

$$\text{In general} \quad P(a < T < b) = e^{-\lambda a} - e^{-\lambda b} \quad \text{where} \quad \lambda = \frac{5}{20} = \frac{1}{4}.$$

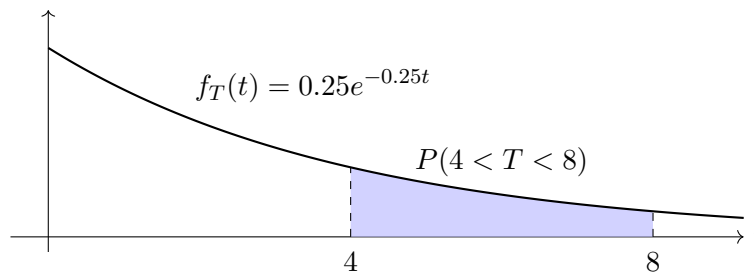
Thus

$$P(4 < T < 8) = e^{-\left(\frac{1}{4}\right)(4)} - e^{-\left(\frac{1}{4}\right)(8)} = e^{-1} - e^{-2} \approx 0.2325.$$

Therefore,

$$P(4 < T < 8) \approx 23.25\%$$

Probability region



9 Monitoring Online Store Orders

[C4-1]

An online store receives customer orders throughout the day. The time between consecutive orders is modeled by an exponential distribution. Based on previous observations, there is a 60% probability that the next order is received within 5 minutes. On average, how many orders are expected to be received in a 12-minute interval?

Questions/Tasks.

- C1** Determine the average number of orders expected to be received in a 12-minute interval. [1]

C4.9 - Number of orders

There is a 60% probability that the next order is received within 5 minutes. Using

$$P(T < t) = 1 - e^{-\lambda t},$$

we obtain

$$0.60 = 1 - e^{-5\lambda} \quad \longrightarrow \quad \lambda = -\frac{\ln(0.40)}{5} \approx 0.1833 \frac{\text{orders}}{\text{minute}}$$

Therefore, over a 12-minute interval, the order rate gives

$$N = (0.1833) \frac{\text{orders}}{\text{minute}} (12 \text{ minutes}) \approx \boxed{2.2 \text{ orders}}$$

10 Waiting for an Economic Indicator Update

[C4-1]

An economic analysis system monitors market indicators throughout the day. The time between consecutive significant indicator updates is modeled by an exponential distribution. Based on previous observations, there is a 25% probability that the next significant update takes more than 10 minutes to occur. Find the average number of significant updates expected per minute.

Questions/Tasks.

C1 Determine the average number of significant updates expected per minute.

[1]

C4.9 - Average number of updates per minute

There is a 25% probability that a significant economic indicator update takes more than 10 minutes to occur. Using

$$P(T > t) = e^{-\lambda t},$$

we obtain

$$0.25 = e^{-10\lambda} \quad \longrightarrow \quad \lambda = -\frac{\ln(0.25)}{10} \approx \boxed{0.1386 \frac{\text{updates}}{\text{minute}}}$$

11 Comparing Economic Forecasting Rates

[C4-2]

Two economic forecasting systems monitor significant changes in market indicators throughout the day. The detected changes by each system are modeled by independent Poisson processes.

For System A, there is a 70% probability that the next significant change is detected within 10 minutes. For System B, there is a 50% probability that the next significant change is detected within 6 minutes.

Questions/Tasks.

C2 Determine after how many minutes the expected numbers of significant changes detected by the two systems differ by 1 change.

[2]

C4.10 - Comparing detection rates

For an exponential waiting time T , if

$$P(T < t) = p \quad \longrightarrow \quad p = 1 - e^{-\lambda t} \quad \longrightarrow \quad \lambda = -\frac{\ln(1 - p)}{t}$$

For System A, $P(T < 10) = 0.70$, so

$$\lambda_A = -\frac{\ln(1 - 0.70)}{10} \approx 0.1204 \frac{\text{changes}}{\text{minute}}$$

For System B, $P(T < 6) = 0.50$, so

$$\lambda_B = -\frac{\ln(1 - 0.50)}{6} \approx 0.1155 \frac{\text{changes}}{\text{minute}}$$

Therefore, System A has the higher detection rate. The difference between the rates is

$$\lambda_A - \lambda_B \approx 0.1204 - 0.1155 = 0.0049 \frac{\text{changes}}{\text{minute}}$$

After t minutes, this rate difference accumulates to $(\lambda_A - \lambda_B)t$ changes. We want this difference to reach 1 change:

$$(\lambda_A - \lambda_B)t = 1. \quad \longrightarrow \quad t = \frac{1}{\lambda_A - \lambda_B} = \frac{1}{0.0049} \approx 204 \text{ minutes}$$

12 Finding a 50% Probability Interval for an Exponential Distribution [CX-1]

The waiting time T between successive economic events is modeled by an exponential distribution with rate $\lambda > 0$. Consider an interval that begins at $t = 1$ minute. The goal is to determine how far the interval must extend so that it contains 50% of the probability represented by the distribution, starting from $t = 1$.

Questions/Tasks.

- C2** For a general rate λ , determine the endpoint b such that the interval $[1, b]$ contains 50% of the probability. Express the length of the interval in terms of λ . Is there any requirement λ must satisfy? [2]

C4.10 - Finding a 50% probability interval

For an exponential random variable with rate λ ,

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0,$$

and its cumulative distribution function is

$$F(t) = P(T \leq t) = 1 - e^{-\lambda t}.$$

Let the interval begin at $t = 1$ and end at $t = b$. Using the cumulative distribution function, we want

$$F(b) - F(1) = P(1 \leq T \leq b) = 0.50.$$

Substituting $F(t) = 1 - e^{-\lambda t}$,

$$(1 - e^{-\lambda b}) - (1 - e^{-\lambda}) = 0.50.$$

Simplifying,

$$e^{-\lambda} - e^{-\lambda b} = 0.50.$$

Therefore,

$$e^{-\lambda b} = e^{-\lambda} - 0.50.$$

Taking the natural logarithm,

$$-\lambda b = \ln(e^{-\lambda} - 0.50) \quad \longrightarrow \quad b = -\frac{1}{\lambda} \ln(e^{-\lambda} - 0.50)$$

Since the interval starts at 1, its length is

$$L = b - 1.$$

Therefore,

$$L = -\frac{1}{\lambda} \ln(e^{-\lambda} - 0.50) - 1.$$

For this interval to contain 50% of the probability, we require

$$e^{-\lambda} > 0.50,$$

which gives

$$\lambda < \ln 2$$